

TrES-5b

R-1940

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_160629 · created 2026-08-06T14:27:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457568.7938 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.054
Transit depth (Rp/Rs)^2	3.42 +/- 1.99 [%]
Orbital Inclination (inc)	89.54 +/- 2.87
Airmass coefficient 1 (a1)	0.79 +/- 0.53
Airmass coefficient 2 (a2)	0.15 +/- 0.82
Scatter in the residuals of the lightcurve fit is	68.52 %
Best Comparison Star	None
Optimal Aperture	0.95
Optimal Annulus	7.11
Transit Duration (day)	0.0872 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.054 vs 0.142 (deviation 0.8 σ)
Duration fit vs published	0.0872 d vs 0.0758 d (deviation 1.65 σ)
Mid-time O–C	-1.6 min
Depth SNR	3.4
Residual scatter	68.52 %

Light curve

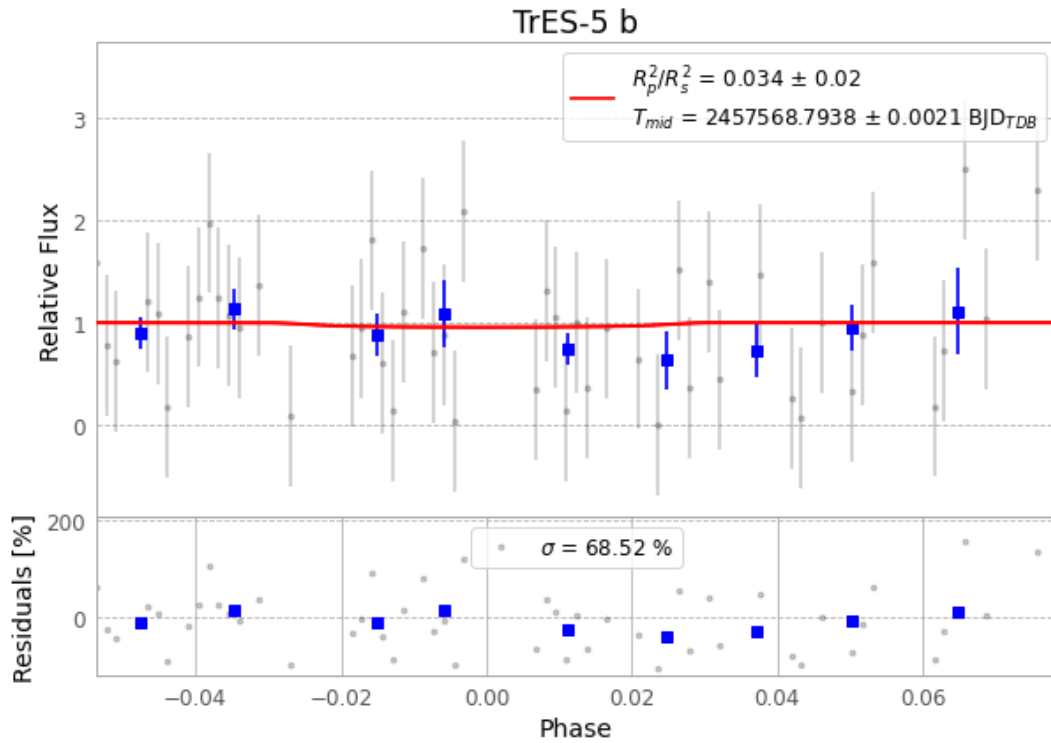


Figure 1 — FinalLightCurve_TrES-5 b_2016-06-28.png (SHA-256 0c58a62a032c2420...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [144, 187], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 057a1a272ec2785c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143

Data provenance

97 FITS frames (64 MB), TRES-5160629050612.FITS → TRES-5160629095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-160629.log (990de072f4fd...), FinalLightCurve_TrES-5 b_2016-06-28.png (0c58a62a032c...), FinalLightCurve_TrES-5 b_2016-06-28.csv (c24847f2d1c9...)

Compute resources

Wall time 198.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:27Z.